Sandeep Malik

AI Architect | Principal Engineer — Enterprise Agent Platforms, Harness Engineering & MCP
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SUMMARY

AI Architect and Principal Software Engineer with 14+ years across AI, data, and distributed systems. I build AI agents that do real engineering work, and the guardrails that make them safe to trust. At Autodesk I lead a team of 4 building the platform other teams use to create and run agents; an agent I built migrated 1,500+ internal applications to GraphQL with over 85% success, with the AI working inside a controlled environment and deterministic checks deciding what ships. I also set how the team builds with AI (shared agentic SDLC workflow, evals and Dynatrace APM monitoring, coaching on tactical vs. strategic programming), and work closely with our Distinguished Architect on platform design. Built on a decade of data platforms: ~200K events/sec search, ~2 PB pipelines, and Databook's first semantic layer.

CORE SKILLS

Agentic AI & Architecture: Enterprise AI solutions, agent harnesses and runtimes, multi-agent orchestration, MCP interoperability, tool integration, context assembly, RAG, deterministic workflows, human oversight
AI Quality & Trust: Agentic SDLC standards, AI-assisted code quality, agent evaluation, LLM-as-judge, guardrails, AI observability (evals, Dynatrace APM, Opik), secure coding, RBAC, vulnerability and compliance data, production monitoring
Models & Frameworks: LangGraph, Google ADK, LangChain, LlamaIndex, Azure OpenAI, AWS Bedrock, Pinecone, Elasticsearch, OpenSearch
Cloud, Data & Engineering: Python, Java, Scala, SQL, Node.js, Snowflake, Databricks, Spark, Kafka, PostgreSQL, AWS, Azure, Kubernetes, Docker, Terraform, CI/CD

EXPERIENCE

Principal Software Engineer / Technical Lead, Autodesk, Trust AI Engineering

Jul 2025 – Present | Montreal, Canada

- Lead a team of 4 building Autodesk's central platform for AI agents and MCP servers (shared catalog, discovery, orchestration, and runtime) so any team can build and run agents safely; work closely with the Distinguished Architect on platform design.
- Migrated 1,500+ internal applications off legacy, vulnerability-prone APIs to GraphQL with an agent I built (Python, Go, Java, Ruby on Rails, Perl) at greater than 85% success by code complexity; the AI works inside a controlled environment and deterministic checks decide what ships, with progress tracked in a UI dashboard.
- Shipped 3 production agents, 5 MCP servers, and 2 IDE plugins used across Autodesk; built a reusable agent runtime (shared auth, typed outputs, guards, observability) so teams assemble agents from common parts.
- Created one shared AI development workflow (intent, design, plan, ADR, spec, implementation loop, docs) packaged as a plugin of skills and agents built on Autodesk guidelines, so every engineer produces the same artifacts.
- Defined how the team monitors agents and MCP servers: evals for output quality and Dynatrace (APM) for runtime health in production; coach engineers on tactical vs. strategic programming with AI and review the quality of AI-generated code.
- Built security into developers' daily tools: an MCP server that checks code against secure-coding and cloud standards in the IDE, a multi-step threat-modeling workflow, and agents that detect repo vulnerabilities and open fix PRs.

Data Architect / Staff Engineer, Databook Labs

Dec 2022 – Jul 2025 | Montreal, Canada (Remote)

- Built the company data platform from the ground up and delivered its first semantic layer (early 2023), powering Databook AI.
- Led a GPT product for strategic sales using multi-query and step-back retrieval, reranking, and LLM-as-judge to improve grounding and personalization.
- Designed agentic workflows with LangGraph and Google ADK and a semantic index across Pinecone, Elasticsearch, Snowflake, and Databricks; led a team of data engineers.

Principal Data Engineer, Sitecore
Mar 2022 – Dec 2022 | Montreal, Canada

- Led the core Search team supporting AI-backed search and e-commerce personalization at ~200,000 events per second.
- Architected a data lake and warehouse for a global retailer, reducing storage costs; delivered Snowflake marts and automated monitoring.

Solutions Architect, Cloudera
May 2021 – Mar 2022 | Montreal, Canada

- Advised customers adopting Cloudera Data Platform across public, private, and hybrid environments; designed on-premises to AWS migrations and trained engineering teams.

Senior Data Engineer, Intact Insurance
Apr 2020 – May 2021 | Montreal, Canada

- Led data engineers delivering an on-premises to AWS historical-data migration and refactoring pipelines onto Databricks; authored cloud and data-platform best-practice guides.

Earlier: Data Engineer, Ericsson (2019–20) · Data Engineer, CGI (2018–19) · Data Engineer, Synechron (2018) · Big Data Developer, Interglobe (2016–18, Dubai — ~2 PB pipelines) · Software Engineer, MothersonSumi/MIND (2014–16) · Software Developer, NIIT (2012–14).

SELECTED PROJECT

Agentic Expense Manager (open source): Local-first Claude tool-calling agent for web and WhatsApp on a shared service layer with a SQLite source of truth (FastAPI, React); server-side tax calculation, receipt OCR, optional Google Drive/Sheets sync; 345+ backend tests including adversarial sync cases.

CERTIFICATIONS

Snowflake SnowPro Core · AWS Solutions Architect – Associate · AWS Cloud Practitioner · Databricks Certified Associate Developer (Spark 3.0) · Cloudera CCA Spark and Hadoop Developer (CCA-175) · Microsoft Azure Fundamentals (AZ-900)

EDUCATION

Master of Science (M.Sc.), Computer Science — Manav Bharti University (2010–2013)